

Harender S. Dhattarwal

PostDoc Associate

Department of Chemistry and Chemical Biology, Rutgers University

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Summary of Qualifications

Experience in the bulk and interfacial molecular dynamic simulations of battery electrolytes. Strong background in classical, quantum, and machine learning potential-based molecular dynamic simulations, including experience in quantitative data management. Skilled in high performance computing.

Education

Ph.D. in Computational Chemistry

Indian Institute of Technology Delhi

PI: Prof. Hemant K. Kashyap (IIT Delhi, India), Co PI: Prof. Jer-Lai Kuo (IAMS, Taiwan)

July 2022

Delhi, India

Master of Science in Chemistry

Indian Institute of Technology Guwahati

July 2017

Guwahati, India

Bachlors of Science in Chemistry

University of Delhi

July 2015

Delhi, India

Research Work

- Mechanistic Insights on Electrolyte Intercalation-deintercalation in Nanoscale Confinements
- Structure and Stability of Electrolytes at the Surface of Metal Anode
- Structure and Dynamics of Ionic Liquids in Bulk and near Interface
- Wetting-Dewetting Behavior of Carbon Surfaces by Ionic Liquids and Deep Eutectic Solvents

Experience

PostDoc Associate

Department of Chemistry and Chemical Biology, Rutgers University

July 2022 -

New Brunswick, NJ, USA

Summer Inter

Institute of Atomic and Molecular Sciences, Academia Sinica

May 2019 - June 2019

Taipei, Taiwan

Winter Inter

Institute of Atomic and Molecular Sciences, Academia Sinica

Oct. 2018 - Nov. 2018

Taipei, Taiwan

Selected Publications

1. **Harender S. Dhattarwal**, Ang Gao, and Richard C. Remsing, Dielectric Saturation in Water from a Long Range Machine Learning Model. *arXiv* **2023**.
2. **Harender S. Dhattarwal** and Hemant K. Kashyap, Heterogeneity and Nanostructure of Superconcentrated LiTFSI-EmimTFSI Hybrid Aqueous Electrolytes: Beyond the 21 m Limit of Water-in-Salt Electrolyte. *J. Phys. Chem. B* **2022** 126 (28), 5291.
3. **Harender S. Dhattarwal**, Jer-Lai Kuo, and Hemant K. Kashyap, Mechanistic Insight on the Stability of Ether and Fluorinated Ether Solvent-Based Lithium Bis (fluoromethanesulfonyl) Electrolytes near Li Metal Surface.

J. Phys. Chem. C **2022** 126 (20), 8953.

4. **Harender S. Dhatarwal** and Hemant K. Kashyap, Unique and Generic Structural Features of Cholinium Amino Acid Biocompatible Ionic Liquids. *Phys. Chem. Chem. Phys.* **2021** 23, 10662.
5. **Harender S. Dhatarwal**, Richard C. Remsing, and Hemant K. Kashyap, Intercalation–deintercalation of water-in-salt electrolytes in nanoscale hydrophobic confinement. *Nanoscale* **2021** 13, 4195.
6. **Harender S. Dhatarwal**, Yun-Wen Chen, Jer-Lai Kuo and Hemant K. Kashyap, Mechanistic Insight on the Formation of Solid Electrolyte Interphase (SEI) by Acetonitrile-Based Superconcentrated [Li][TFSI] Electrolyte Near Li Metal. *J. Phys. Chem. C* **2020** 124, 27495.
7. **Harender S. Dhatarwal**, Richard C. Remsing and Hemant K. Kashyap, How Flexibility of Nanoscale Solvophobic Confining Material Promotes Capillary Evaporation of Ionic Liquids. *J. Phys. Chem. C* **2020** 124, 4899.
8. Akshay Malik, **Harender S. Dhatarwal** and Hemant K. Kashyap, Distinct Solvation Structures of CO₂ and SO₂ in Reline and Ethaline Deep Eutectic Solvents Revealed by AIMD Simulations. *J. Phys. Chem. B* **2021** 125, 1852.

Awards

- PCCP Best Poster Presentation Award at Theoretical Chemistry Symposium (TCS 2021) IISER Kolkata, Kolkata.
- Best Poster Presentation Award at Advanced Simulation Methods: DFT, MD and Beyond (ASM 2019), IIT Delhi, Delhi.
- Senior Research Fellowship awarded by UGC-India (July 2019–Till date).
- Junior Research Fellowship awarded by UGC-India (July 2017–July 2019).

Skills

Programming	Fortran, Python, C++, Bash scripting
Scientific	GROMACS, LAMMPS, GAUSSIAN, CP2K
Graphics	Inkscape, VMD, Inkscape, Xmgrace, Avogadro, Gnuplot, Matlab, Gauss-view